

17-1d How the Size of an Oligopoly Affects the Market Outcome

We can use the insights from this analysis of duopoly to discuss how the size of an oligopoly is likely to affect the outcome in a market. Suppose, for instance, that John and Joan suddenly discover water sources on their property and join Jack and Jill in the water oligopoly. The demand schedule in [Table 1](#) remains the same, but now more producers are available to satisfy this demand. How would an increase in the number of sellers from two to four affect the price and quantity of water in the town?

If the sellers of water could form a cartel, they would once again try to maximize total profit by producing the monopoly quantity and charging the monopoly price. Just as when there were only two sellers, the members of the cartel would need to agree on production levels for each member and find some way to enforce the agreement. As the cartel grows larger, however, this outcome is less likely. Reaching and enforcing an agreement becomes more difficult as the size of the group increases.

If the oligopolists do not form a cartel—perhaps because the antitrust laws prohibit it—they must each decide on their own how much water to produce. To see how the increase in the number of sellers affects the outcome, consider the decision facing each seller. At any time, each well owner has the option to raise production by one gallon. In making this decision, the well owner weighs the following two effects:

- *The output effect:* Because price is above marginal cost, selling one more gallon of water at the going price will raise profit.
- *The price effect:* Raising production will increase the total amount sold, which will lower the price of water and lower the profit from all the other gallons sold.

If the output effect outweighs the price effect, the well owner will increase production. If the price effect outweighs the output effect, the owner will not raise production. (In fact, in this case, it is profitable to reduce production.) Each oligopolist increases production until these two marginal effects exactly balance, taking the other firms' production as given.

Now consider how the number of firms in the industry affects the marginal analysis of each oligopolist. The more sellers there are, the less each seller is concerned about her own impact on the market price. That is, as the oligopoly grows in size, the magnitude of the price effect falls. When the oligopoly grows very large, the price effect disappears altogether. In this extreme case, the production decision of an individual firm no longer affects the market price. Each firm takes the market price as given when deciding how much to produce and, therefore, increases production as long as price exceeds marginal cost.

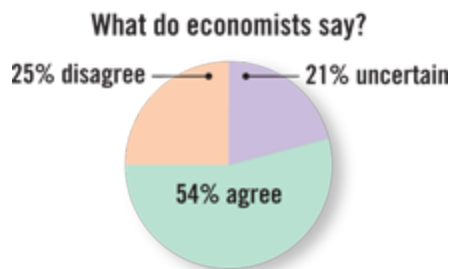
We can now see that a large oligopoly is essentially a group of competitive firms. A competitive firm considers only the output effect when deciding how much to produce: Because a competitive firm is a price taker, the price effect is absent. Thus, *as the number of sellers in an oligopoly grows, an oligopolistic market increasingly resembles a competitive market. The price approaches marginal cost, and the quantity produced approaches the socially efficient level.*

This analysis of oligopoly offers a new perspective on the effects of international trade. Imagine that Toyota and Honda are the only automakers in Japan, Volkswagen and BMW are the only automakers in Germany, and Ford and General Motors are the only automakers in the United States. If these nations prohibited international trade in autos, each would have an auto oligopoly with only two members, and the market outcome would likely depart substantially from the competitive ideal. With international trade, however, the car market becomes a world market, and the oligopoly in this example has six members. Allowing free trade increases the number of producers from which each consumer can choose, and this increased competition keeps prices closer to marginal cost. Thus, the theory of oligopoly provides another reason, in addition to the theory of comparative advantage discussed in [Chapter 3](#), why countries can benefit from free trade.

Ask the Experts

Market Share and Market Power

“If a small number of firms have a large combined market share in a properly defined market, it is strong evidence that those firms have substantial market power.”



Source: IGM Economic Experts Panel, September 25, 2018.

QuickQuiz

1. The key feature of an oligopolistic market is that
 - a. each firm produces a different product from other firms.

- b. a single firm chooses a point on the market demand curve.
 - c. each firm takes the market price as given.
 - d. a small number of firms are acting strategically.
2. If an oligopolistic industry organizes itself as a cooperative cartel, it will produce a quantity of output ____ the competitive level and ____ the monopoly level.
- a. less than; more than
 - b. more than; less than
 - c. less than; equal to
 - d. equal to; more than
3. If an oligopoly does not cooperate and each firm chooses its own quantity, the industry will produce a quantity of output ____ the competitive level and ____ the monopoly level.
- a. less than; more than
 - b. more than; less than
 - c. less than; equal to
 - d. equal to; more than
4. As the number of firms in an oligopoly grows, the industry approaches a level of output ____ the competitive level and ____ the monopoly level.
- a. less than; more than
 - b. more than; less than
 - c. less than; equal to
 - d. equal to; more than

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